
INTRODUCTION TO CIVIL ENGINEERING

Course Code: ESC131/231

Credits: 3:0:0

Pre-requisites: Nil

Contact Hours: 42L

Course Coordinator: Dr. B Suguna Rao

Course Content

Unit I

Civil Engineering Disciplines and Building Science

Introduction to Civil Engineering: Surveying, Structural Engineering, Geotechnical Engineering, Hydraulics & Water Resources, Transportation Engineering, Environmental Engineering, Construction planning & Project management.

Basic Materials of Construction: Bricks, Cement & mortars, Plain, Reinforced & Pre-stressed Concrete, Structural steel, Construction Chemicals.

Structural elements of a building: Foundation, plinth, Masonry wall, column, beam, slab, and staircase.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Links: Introduction: <https://a.impartus.com/ilc/#/video/id/534326>
- Scopes of Various fields: <https://a.impartus.com/ilc/#/video/id/536441>

Self Study: Building components such as Lintel and Chejja

Unit II

Societal and Global Impact of Infrastructure

Infrastructure: Introduction to sustainable development goals, Smart city concept, Safe city concept

Environment: Water Supply and Sanitary systems, urban air pollution management, Solid waste management, urban flood control

Built-environment: Energy efficient buildings, recycling, Temperature and Sound control in buildings, Security systems; Smart buildings.

- Pedagogy/ Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Links: Infrastructure Habitats: <https://www.youtube.com/watch?v=wpvbVyUCi78>
<https://www.youtube.com/watch?v=Irq4BigInDU>

Self Study: Clean city concept, identification of Landfill sites.

Unit III

Analysis of force systems: Concept of idealization, system of forces, principles of superposition and transmissibility, Resolution and composition of forces, Resultant of concurrent and non-concurrent coplanar force systems, moment of forces, couple, Varignon's theorem, free body diagram, equations of equilibrium, equilibrium of concurrent and non-concurrent coplanar force systems.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Links: Introduction to:
Engg. Mechanics - <https://a.impartus.com/ilc/#/video/id/532408>
Resultants - <https://a.impartus.com/ilc/#/video/id/537603>
Equilibrium Concepts: <https://a.impartus.com/ilc/#/video/id/550330>

Self Study: Law of Parallelogram of forces, Numerical on Couple, and moment.

Unit IV

Centroid: Importance of centroid and centre of gravity, methods of determining the centroid, locating the centroid of plane laminae from first principles, centroid of built-up sections. Numerical examples.

Friction: Definition of friction and its application, angle of friction, angle of repose, coefficient of friction, Types of friction, laws of static friction, Description, and application of friction on blocks on horizontal and inclined planes.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Links: https://www.youtube.com/watch?v=QK_TuE2lfSc
- Friction concepts: <https://www.youtube.com/watch?v=AlenFWwK3Ek>

Self Study: Derivation of Centroid of semicircle and quarter circle.

Unit V

Moment of inertia: Importance of Moment of Inertia, method of determining the second moment of area (moment of inertia) of plane sections from first principles, parallel axis theorem and perpendicular axis theorem, section modulus, radius of gyration, moment of inertia of built-up sections, Numerical Examples.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations
- Links: Moment of Inertia: <https://www.youtube.com/watch?v=Bl5KqQOWkY>

Self Study: Derivation of moment of inertia of semicircle and quarter circle.

Text Books:

1. **Bansal R. K., Rakesh Ranjan Beohar and Ahmad Ali Khan** - Basic Civil Engineering and Engineering Mechanics, Laxmi Publications, 2015.
2. **Kolhapure B K** - Elements of Civil Engineering and Engineering Mechanics, EBPB, 2014.

Reference Books:

1. **Beer F.P. and Johnston E. R.** - Mechanics for Engineers, Statics and Dynamics, McGraw Hill, 1987.
2. **Irving H. Shames** - Engineering Mechanics, Prentice-Hall, 2019.
3. **Hibbler R. C.** - Engineering Mechanics: Principles of Statics and Dynamics, Pearson Press, 2017.
4. **Timoshenko S, Young D. H., Rao J. V.** - Engineering Mechanics, Pearson Press, 5th edition, 2017.

5. **Bhavikatti S S** - Engineering Mechanics, New Age International, 2019.
6. **Reddy Vijaykumar K and Suresh Kumar K**, Engineering Mechanics, BS publication, 2011.

Web links and video Lectures (e- Resources):

1. <https://www.youtube.com/watch?v=nGfVTNfNwnk&list=PLOSWwFV98rfKXq2KBphJz95rao7q8PpwT>
2. <https://www.youtube.com/watch?v=nkg7VNW9UCc&list=PLOSWwFV98rfKXq2KBphJz95rao7q8PpwT&index=2>
3. <https://www.youtube.com/watch?v=3YBXteL-qY4>
4. <https://www.youtube.com/watch?v=z95UW4wwzSc&list=PLOSWwFV98rfKXq2KBphJz95rao7q8PpwT&index=10>
5. <https://www.youtube.com/watch?v=lheoBL2QaQU&list=PLOSWwFV98rfKXq2KBphJz95rao7q8PpwT&index=7>
6. https://www.youtube.com/watch?v=atoP5_DeTPE
7. <https://www.youtube.com/watch?v=kmsmp9OzAsI>
8. <https://www.youtube.com/watch?v=x1ef048b3CE>

Course Outcomes (COs):

On successful completion of the course students will be able to

1. Understand the vast interface of civil engineering fields towards the society at large.
2. Recognize the societal and global impact towards employment creation and its contribution to the GDP.
3. Analyze and evaluate the coplanar concurrent and non concurrent force system.
4. Locate the centroid of bodies and frictional force developed between the bodies.
5. Compute the moment of inertia of plane areas about a given axis.

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tool	Marks	Course outcomes attained
Internal Assessment-I	30	CO1, CO3, CO4
Internal Assessment-II	30	CO2, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Assignment	10	CO1, CO2
Assignment	10	CO3, CO4, CO5
Semester End Examination (SEE)		
Course end examination (Answer one question from each unit –internal choice)	100	CO1, CO2, CO3, CO4, CO5